

Neal Miller

ENGINEERING LEADER · AI-FIRST ENGINEERING · PRODUCT DELIVERY · ARCHITECTURAL TRANSFORMATION & MODERNIZATION

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Summary

Engineering leader with experience building and scaling multi-team organizations across complex, high-stakes product environments. Currently overseeing five engineering teams at Imagine Learning — spanning Imagine Learning’s market-leading courseware business and a newly founded platform engineering team — with a focus on modernization, cross-product infrastructure, and long-term technical strategy. A practitioner and organizational champion of agentic, AI-first development, leading one of the earliest end-to-end adoptions of this model in a commercial engineering organization. Ph.D., Columbia University.

Core Competencies

Leadership & Strategy

Organizational leadership, Multi-team management, Technical strategy, Product alignment, Hiring & talent development, Vendor/contract management, Agile & PI planning, AI-first SDLC transformation, Org design, Portfolio-level prioritization, Change management

Technical Expertise

Service-oriented architecture (SOA), Event-driven systems, Distributed systems, Scalable web applications, System design, Data-intensive applications, Cloud-native modernization, Architectural governance

AI & Engineering Practice

Agentic development workflows, AI-augmented delivery, LLM tooling & integration, Engineering productivity, Claude Code / GitHub Copilot

Experience

Senior Software Engineering Manager

Remote

IMAGINE LEARNING

June 2025 - Present

Lead engineering across Imagine Learning’s product portfolio — currently overseeing **EdgeEX**, the next-generation courseware platform serving ~**400K** high school students and the strategic successor to Imagine Learning’s legacy Edgenuity platform, across 4 teams of ~24 engineers; and founding and leading a new **Learning Platform** team; previously led the **Literacy Suite** (3 teams, ~20 engineers).

Leadership & Delivery

- Lead ~27 FTEs and contractors across 5 teams, with ~8 direct reports.
- Drive cross-team architectural alignment, dependency management, and delivery execution across multiple product areas.
- Partner with Product, Design, and Curriculum leadership to define roadmap priorities and deliver high-impact, user-centered solutions.

Technical & Organizational Impact

- Founded the Learning Platform team to build shared platform services (Calendar, Assignment, Gradebook, Enrollment, Slide Player) serving all Imagine Learning products, with an initial focus on powering a new high school science product targeting a Fall 2026 pilot.
- Established the Learning Platform team as Imagine Learning’s first adopter of an agentic, AI-first development model — a company-wide transformation spanning the full product lifecycle from design and requirements through delivery.
- Champion adoption of AI-assisted development tools (Claude Code, GitHub Copilot) to improve productivity and code quality across engineering.
- Architect and oversee a multi-year effort to modernize legacy monoliths, reducing operational risk and enabling faster product delivery across multiple teams.

Software Engineering Manager

Remote

IMAGINE LEARNING

June 2023 - June 2025

Led the Literacy Data & Insights team, delivering real-time educator-facing reports within the Literacy Suite, which serves **1M+** students.

- Delivered the Growth Report for BTS 2024 — the most technically complex report in the Literacy Suite — spanning district, school, group, and student views on a new real-time data architecture.

- Managed a team of up to ~8 software engineers and SDETs (mix of FTEs and contractors).
- Ensured alignment through quarterly PI planning, managing cross-team dependencies and engineering commitments.
- Led hiring for both FTEs and contractors and managed vendor relationships.

Software Engineering Manager

Remote

SMILEDIRECTCLUB

June 2022 - June 2023

Led the Core Services team, building foundational internal services supporting user-facing products across SmileDirectClub's engineering organization, including contributing infrastructure supporting **SmileOS** — the company's AI-powered treatment planning platform serving 1.5M+ customers.

- Managed a blended team of ~6 engineers and contractors.
- Drove technical alignment and prioritization across product, business, and engineering stakeholders.
- Served as the primary liaison between Core Services and the broader organization, ensuring clarity of communication and strategic alignment.

Software Engineering Team Lead

Remote

SMILEDIRECTCLUB

August 2020 - June 2022

- **Partner Network**
 - Transformed team delivery practices — eliminating dysfunctional sprint ceremonies and transitioning from multi-hour nightly releases to on-demand deployments with minimal overhead.
 - Led development of the Partner Portal supporting SDC's Partner Network initiative.
- **Prescriptions**
 - Led the transition of the Prescriptions domain from a monolithic Django application to an event-driven, service-oriented architecture.
 - Delivered the Treatment Plan Service and shifted all functionality with zero downtime and minimal disruption.

Senior Lead Software Developer

Houston, TX

FRANCHICZAR

March 2019 - August 2020

Led technical direction for a 6-person development team, driving architecture, delivery, and hands-on development across multiple products.

- Played a central role in hiring and onboarding of engineering team members.
- Oversaw development of multiple products, including:
 - FranchiCzar OS – CRM for franchise-based sales
 - Fund.io – payment and subscription management platform
 - Rylos – learning management system (LMS)

Software Engineer Team Lead

Houston, TX

GENE BY GENE

November 2017 - March 2019

Senior Software Developer

Houston, TX

DURWELLA

February 2016 - November 2017

Lead Software Engineer

Houston, TX

RIGMINDER

June 2014 - February 2016

Learning Analytics Engineer

Houston, TX

REASONING MIND

July 2012 - June 2014

Developed algorithms for educational data mining within the Reasoning Mind adaptive learning platform, co-authoring peer-reviewed research on automated detection of student engagement and teacher behavior across a cohort of 70,000 students.

Education

Columbia University in the City of New York

New York, NY

PH.D. CHEMICAL PHYSICS (WITH DISTINCTION)

2012

- Dissertation: *Self-organization in systems of anisotropic particles*

The University of Texas at Austin

Austin, TX

B.S. CHEMISTRY, HONORS (WITH HIGHEST HONORS)

2007